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Computer Science & Engineering

Experiment 9

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Subject Name: Database
Management System

Subject Code: 24CSH-298

AIM: This experiment focuses on designing and implementing a PL/SQL Package in Oracle. It demonstrates how to use a package specification and package body, along with procedures and shared cursors, to achieve modular and reusable database programming.

OBJECTIVES:

- To understand PL/SQL packages
- To learn package specification and package body
- To implement procedures inside packages
- To use shared cursors within a package
- To execute package procedures and display results

SOFTWARE REQUIREMENTS:

- Oracle Database / SQL*Plus / Oracle Live SQL
- SQL (DDL, DML)
- PL/SQL

PROCEDURE/STEPS:

1. Start the system and log in
2. Open Oracle SQL*Plus / Live SQL



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3. Connect to the database
4. Create employees table
5. Insert sample data
6. Create package specification
7. Create package body with procedures and cursor
8. Execute package procedures
9. Observe output

PRACTICAL/EXPERIMENTAL STEPS:

```
CREATE TABLE employees (  
    emp_id NUMBER PRIMARY KEY,  
    emp_name VARCHAR2(50),  
    salary NUMBER  
);
```

```
INSERT INTO employees VALUES (1, 'Anhad', 30000);  
INSERT INTO employees VALUES (2, 'Jaskaran', 40000);  
INSERT INTO employees VALUES (3, 'Yuvraj', 50000);
```

```
COMMIT;
```

```
CREATE OR REPLACE PACKAGE emp_package AS  
    PROCEDURE show_employees;  
    PROCEDURE get_employee(p_id NUMBER);  
END emp_package;  
/
```

```
CREATE OR REPLACE PACKAGE BODY emp_package AS
```

```
    CURSOR emp_cursor IS  
        SELECT emp_id, emp_name, salary FROM employees;
```

```
    PROCEDURE show_employees IS  
    BEGIN  
        FOR rec IN emp_cursor LOOP  
            DBMS_OUTPUT.PUT_LINE(  
                'ID: ' || rec.emp_id ||  
                ' Name: ' || rec.emp_name ||
```



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```
        ' Salary: ' || rec.salary
    );
END LOOP;
END;

PROCEDURE get_employee(p_id NUMBER) IS
BEGIN
    FOR rec IN emp_cursor LOOP
        IF rec.emp_id = p_id THEN
            DBMS_OUTPUT.PUT_LINE(
                'Employee Found -> ID: ' || rec.emp_id ||
                ' Name: ' || rec.emp_name ||
                ' Salary: ' || rec.salary
            );
        END IF;
    END LOOP;
END;

END emp_package;
/

SET SERVEROUTPUT ON;

BEGIN
    emp_package.show_employees;
END;
/

BEGIN
    emp_package.get_employee(2);
END;
/
```

OUTPUT:(Screenshots)



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- Displays all employee records using show_employees

```
1 DROP TABLE IF EXISTS employees;
2 CREATE TABLE employees (
3     emp_id NUMBER PRIMARY KEY,
4     emp_name VARCHAR2(50),
5     salary NUMBER
6 );
7
8 INSERT INTO employees VALUES (1, 'Anhad', 30000);
9 INSERT INTO employees VALUES (2, 'Jaskaran', 40000);
10 INSERT INTO employees VALUES (3, 'Yuvraj', 50000);
11
12 COMMIT;
13
14 CREATE OR REPLACE PACKAGE emp_package AS
```

Query result Script output **DBMS output** Explain Plan SQL history

ID: 1 Name: Anhad Salary: 30000
ID: 2 Name: Jaskaran Salary: 40000
ID: 3 Name: Yuvraj Salary: 50000

Employee Found -> ID: 2 Name: Jaskaran Salary: 40000

- Displays specific employee details using get_employee

```
1 DROP TABLE IF EXISTS employees;
2 CREATE TABLE employees (
3     emp_id NUMBER PRIMARY KEY,
4     emp_name VARCHAR2(50),
5     salary NUMBER
6 );
7
8 INSERT INTO employees VALUES (1, 'Anhad', 30000);
9 INSERT INTO employees VALUES (2, 'Jaskaran', 40000);
10 INSERT INTO employees VALUES (3, 'Yuvraj', 50000);
11
12 COMMIT;
13
14 CREATE OR REPLACE PACKAGE emp_package AS
```

Query result **Script output** DBMS output Explain Plan SQL history

PL/SQL procedure successfully completed.
Elapsed: 00:00:00.124

SQL> BEGIN
emp_package.get_employee(2);
END;

Employee Found -> ID: 2 Name: Jaskaran Salary: 40000

PL/SQL procedure successfully completed.
Elapsed: 00:00:00.004



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I/O ANALYSIS:

Input:

- Employee ID (for specific search)

Output:

- List of all employees
- Details of a specific employee
- Employee ID, Name, and Salary

LEARNING OUTCOMES:

- Understood PL/SQL package structure
- Learned difference between package specification and body
- Implemented procedures inside packages
- Used shared cursor for efficient data handling
- Executed package procedures successfully

RESULT:

The PL/SQL package was successfully created and executed. It demonstrated modular programming using procedures and shared cursors, improving code reusability and efficiency in database operations.